

Coordinate-Invariant Spatial Modeling for Gene Expression Prediction from Histology

Anonymous authors

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Abstract

Predicting spatial gene expression from routine histology images could extend molecular analysis to tissue samples without spatial transcriptomic measurements. This task requires integrating local morphology with tissue context while avoiding sensitivity to arbitrary choices of coordinate origin and orientation. We introduce MIST, a compact spatial transformer that combines three sources of context in each layer: attention over nearby spots, attention over all spots on a slide, and a pooled slide representation. The local pathway uses continuous pairwise distances to bias attention, retaining information about spatial proximity beyond a binary neighborhood graph. Because coordinates enter the model only through these distances, predictions are invariant to translations, rotations, and reflections of the coordinate frame when image features are fixed. We evaluate MIST on seven human organ cohorts from HEST-1k through intra-cohort prediction, pooled multi-organ testing on held-out samples, and leave-one-organ-out cross-organ transfer, using matched features, gene panels, splits, and model-selection procedures within each comparison. In intra-cohort prediction of 100 genes, MIST improves mean Pearson correlation by 11.4% over the strongest baseline and leads on all six cohorts. In pooled evaluation on held-out samples from represented organs, it improves correlation by 0.3% over the strongest baseline. In a separate 50-gene cross-organ evaluation with an entire organ held out during training, MIST improves correlation by 3.6% over BLEEP, the strongest baseline considered. Across frozen UNI, CONCH, and combined UNI + CONCH features, MIST consistently achieves the highest mean correlation among the methods in the pooled encoder comparison. Factorial ablations identify local attention as the main contributor to predictive accuracy. With 4.88M parameters and model-only inference of 64.7 ms for 10,000 spots on an A100, MIST supports accurate, efficient prediction across cohorts and feature encoders, while generalization to unseen organs remains a challenge.

1 Introduction

Spatial transcriptomics (ST) measures gene expression together with tissue location, enabling the study of spatial heterogeneity, tissue organization, and cellular interactions [Ståhl et al. \(2016\)](#). Its cost and experimental burden, however, constrain the number and diversity of specimens that can be profiled. In contrast, H&E slides are routinely collected and contain morphological signals associated with tissue composition and molecular state. Predicting spatial gene expression from these images could therefore extend molecular analysis to specimens without matched ST measurements.

Morphology and expression interact locally, across the slide, and at the level of slide composition. Existing predictors capture parts of this structure with patch regression, spatial graphs, retrieval, or conditional generation, but absolute coordinates can make predictions depend on an arbitrary scanner frame. We therefore seek a compact predictor that combines multi-scale context while using coordinates only through relative distances.

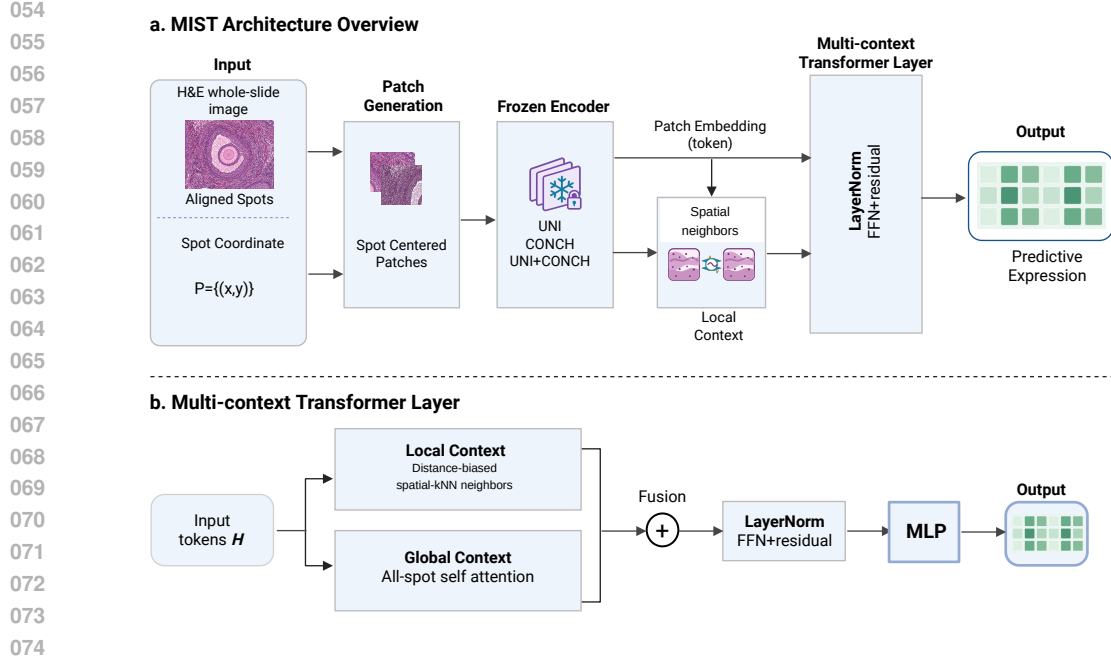


Figure 1: MIST overview. (a) Frozen UNI, CONCH, or combined features turn H&E spot patches into tokens, while coordinate-based k NN edges provide local context. (b) Each layer combines distance-biased local attention with global self-attention before the expression head. Pairwise-distance coordinates ensure invariance to translations, rotations, and reflections.

Table 1: Architectural components across representative H&E→ST predictors. MIST is the only model that combines local neighborhood attention, slide-wide global attention, a slide-level summary token, coordinate invariance, and continuous (distance-expressive) spatial encoding rather than binary adjacency. • = present, ○ = absent.

Method	Local context	Global context	Slide token	Coord. invariant	Cont. distance
ST-Net He et al. (2020)	○	○	○	○	○
HisToGene Pang et al. (2021)	○	•	○	○	○
Hist2ST Zeng et al. (2022)	•	•	○	○	○
BLEEP Xie et al. (2023)	○	○	○	○	○
TRIPLEX Chung et al. (2024)	•	•	○	○	○
M2OST Wang et al. (2025)	•	○	○	○	○
STEM Zhu et al. (2025)	○	•	○	○	○
MIST (ours)	•	•	•	•	•

We introduce MIST, a direct spatial transformer that represents each spot as a token and combines distance-biased k NN attention for local context, full self-attention for slide-wide context, and attention pooling for a slide summary. Continuous radial-basis distance features preserve inter-spot geometry while guaranteeing invariance to translations, rotations, and reflections when image features are fixed (Fig. 1).

We evaluate matched features, panels, splits, budgets, and nested validation in intra-cohort, pooled cross-organ, and leave-one-organ-out (LOOO) settings. Across six human cohorts, MIST reaches mean PCC 0.496 versus 0.444 for BLEEP, 0.760 pooled PCC, and 0.490 LOOO PCC; local attention is the strongest factorial component. The model has 4.88M parameters and processes 10,000 spots in 64.7 ms on an A100.

We investigate five questions: RQ1 (Performance) accuracy across diverse organs, both per-organ and with a single pooled model; RQ2 (Transfer) generalization to an entirely unseen organ under LOOO; RQ3 (Design) the effect of encoder, context streams, and neighborhood

definition; RQ4 (Structure) spatial-domain recovery; and RQ5 (Efficiency) whole-slide inference.

Contributions.

1. We introduce MIST, a multi-scale transformer with local, slide-wide, and slide-summary pathways driven by continuous pairwise-distance features and coordinate-frame invariance.
2. We quantify the effects of context, distance encoding, and feature choice through matched ablations and transformation tests.
3. We provide a controlled pooled cross-organ benchmark alongside intra-cohort and held-out-organ evaluation with matched panels, splits, baselines, and model selection.

2 Related Work

H&E→ST prediction. Prior models span patch-level regression (ST-Net [He et al. \(2020\)](#)), graph spatial context (Hist2ST [Zeng et al. \(2022\)](#)), cross-modal retrieval (BLEEP [Xie et al. \(2023\)](#)), and conditional generation (STEM [Zhu et al. \(2025\)](#)). Because they differ in features, gene panels, splits, and selection, published scores conflate data choices with architecture; we therefore compare under a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometry and generalization. Absolute position encodings can change when an equivalent tissue layout is translated, rotated, or reflected; geometric networks instead build interactions from relative quantities [Satorras et al. \(2021\)](#). MIST conditions its spatial pathway only on pairwise Euclidean distances, invariant to such frame changes for fixed histology features. For generalization we separate pooled cross-validation (organs seen, samples held out) from LOOO (an entire organ withheld) to isolate anatomical transfer on HEST-1k [Jaume et al. \(2024\)](#). An extended discussion and positioning of MIST are in Appendix B.

3 Method

MIST maps frozen spot-level histology features and measured spot coordinates to a spatial gene expression map. Its method has three stages: coordinate-invariant distance encoding, a shared multi-context layer, and supervised prediction with a correlation-aware objective.

Core Idea: MIST predicts a slide’s spatial expression in a single forward pass by fusing three views of context—local distance-biased neighborhoods, global self-attention, and a slide-level token—over frozen histology features, made invariant to the spot coordinate frame (translation, rotation, reflection) so a single model transfers across slides without per-dataset retraining.

3.1 Problem Setup

For a slide with N spots, let $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times d_{\text{in}}}$ denote frozen pathology features and $\mathbf{P} = [\mathbf{p}_1, \dots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$ the corresponding spot coordinates. The target $\mathbf{Y} \in \mathbb{R}^{N \times G}$ contains log-normalized expression for G genes. We learn $f_\theta : (\mathbf{F}, \mathbf{P}) \mapsto \hat{\mathbf{Y}}$ and process each slide independently; the model therefore supports a variable number of spots and does not require expression measurements at test time. In the feature-matched main comparisons, $\mathbf{f}_i$ concatenates frozen UNI [Chen et al. \(2024\)](#) and CONCH [Lu et al. \(2024\)](#) embeddings ($d_{\text{in}} = 1536$).

3.2 Coordinate-Invariant Distance Encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i , we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbors, excluding the spot

itself. Distances are expanded into M radial-basis features,

$$\phi(D_{ij}) = [\exp(-\gamma_m D_{ij}^2)]_{m=1}^M, \quad (1)$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

Proposition 1 (Coordinate frame invariance):

Let $R \in \mathbb{R}^{2 \times 2}$ be orthogonal, let $\mathbf{t} \in \mathbb{R}^2$, and define $\mathbf{P}' = \mathbf{P}R^\top + \mathbf{1}\mathbf{t}^\top$. For fixed histology features $\mathbf{F}$, the prediction is unchanged:

$$f_\theta(\mathbf{F}, \mathbf{P}') = f_\theta(\mathbf{F}, \mathbf{P}). \quad (2)$$

Thus, the spatial pathway is invariant to translations, rotations, and reflections of the coordinate frame in exact arithmetic. The guarantee concerns a change of coordinate frame with fixed histology features; it does not claim that the frozen image encoder is invariant to transformations of the histology image. The proof is in Appendix M. The RBF bias is available to weight neighbors by distance, but at native HEST pixel scale the neighbor distances ($\sim 10^3$) saturate $\exp(-\gamma D^2)$ toward zero for all bandwidths, so in the reported configuration the bias contributes negligibly and the local pathway reduces to attention over the spatial k NN neighborhood. The distinguishing property of MIST is therefore this spatial locality—each spot aggregates from its geometric neighbors rather than from arbitrary or random spots—which our ablation isolates directly (spatial k NN vs. random neighbors, §5.3). The invariance guarantee above holds regardless, since it depends only on distances being rotation-invariant.

3.3 Multi-Context MIST Layer

Core Idea. At every layer, each spot receives three messages in parallel: one from its nearby spots, one from all spots, and one from a pooled summary of the slide. The model adds these messages to the current spot representation and refines the result with a feed-forward network. Only the nearby-spot message uses coordinates, and it uses pairwise distances rather than absolute positions.

The input features are projected to hidden width d , $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbor graph and radial-basis features are reused in all L transformer layers. At layer ℓ , a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

3.3.1 Local distance-biased attention

Each spot attends to its k neighbors with a learned, distance-dependent bias for each attention head. For head h , the attention weights are

$$\alpha_{ij}^h = \text{softmax}_{j \in \mathcal{N}(i)} \left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij})) \right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query-key term, while spatial distance contributes through the learned radial-basis bias.

3.3.2 Global self-attention

In parallel, standard multi-head self-attention Vaswani et al. (2017) allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

3.3.3 Attention-pooled slide context

A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g \left(\sum_{i=1}^N a_i \mathbf{h}_i \right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide’s spot features and summarizes its morphological composition.

For the complete model, the local ($\mathbf{L}$), global ($\mathbf{G}$), and broadcast slide ($\mathbf{S}$) representations are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\tilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{G}^{(\ell)} + \mathbf{S}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \tilde{\mathbf{X}}^{(\ell)} + \text{FFN}(\text{LN}(\tilde{\mathbf{X}}^{(\ell)})). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

Proposition 2 (Permutation equivariance):

Let $Q \in \{0, 1\}^{N \times N}$ be a permutation matrix. If the k NN construction handles distance ties equivariantly, then

$$f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P}). \quad (7)$$

Therefore, reordering the input spots only reorders their predicted expression profiles. Appendix M provides the proof. Our numerical checks evaluate both Propositions 1 and 2 under finite precision.

3.4 Training Objective and Ablations

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda \left(1 - \frac{1}{|\mathcal{V}|} \sum_{g \in \mathcal{V}} \rho_g \right), \quad (8)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

The main model enables all three context pathways. Our factorial ablation independently toggles the local (L), global (G), and slide-level (S) pathways, avoiding conclusions tied to a particular order of architectural additions.

4 Experimental Setup

We evaluate on HEST-1k [Jaume et al. \(2024\)](#) with frozen UNI+CONCH patch features and $\log(1+x)$ targets; unless stated otherwise the target is the 50-gene HEST HVG panel. Cross-organ experiments use the six retained cohorts of Appendix A (HCC, LYMPH_IDC, and READ excluded). We study three regimes: intra-cohort (a model per organ), pooled (train across organs, hold out slide-level samples; organ identity is not a model input), and LOOO (train on all-but-one organ, test the held-out organ). We report PCC—and MSE/MAE where noted—over outer folds and three seeds, with checkpoints chosen on inner validation and evaluated once on the outer fold. Baselines ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) use matched features,

panels, splits, budgets, and selection. MIST has 4.88M parameters (four layers, width 256, eight neighbors). Full dataset, gene-panel, protocol, and hyperparameter details are in Appendix C.

5 Results

5.1 RQ1: Predictive performance across diverse organs

We first establish accuracy on organs the model has seen, both when a model is trained per cohort (intra-cohort) and when a single model is trained jointly across organs (pooled). Table 3 reports the feature-matched comparison across gene-panel sizes. For PCC@50, MIST obtains the highest mean PCC (0.496), compared with 0.444 for BLEEP, the strongest baseline, an 11.6% relative improvement. It leads on all six cohorts, though the margin is not uniform: it is widest on kidney and prostate (+0.072 and +0.064 over the best baseline) and narrowest on lung (+0.024). The same ranking largely holds under a smaller 10-gene panel (PCC@10 in Table 3): with everything else fixed and only the panel reduced to the ten highest-variance genes of each cohort’s HEST 50-HVG set, MIST again leads the average (0.600 vs. 0.545 for BLEEP) and tops five of six cohorts. Absolute correlations rise across all methods because the retained genes are the most variable, and hence the most predictable.

Table 4: Error metrics across gene-selection strategies and panel sizes on HEST-1k. Mean over three seeds, averaged over the same six cohorts as Tables 2 and 3 (IDC, CCRCC, LUNG, PAAD, PRAD, SKCM); lower values are better and the best result in each row is in bold. Models, features, splits, and training budget are matched.

Section	Panel	MSE ↓					MAE ↓				
		ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
Gene-selection strategy	HVG	1.995	1.561	1.318	1.747	1.233	0.969	0.890	0.819	0.920	0.794
	DEG	3.336	1.757	1.557	1.958	1.477	1.457	1.021	0.945	1.054	0.928
	HMHVG	3.589	1.832	1.566	1.976	1.393	1.470	1.042	0.936	1.049	0.901
Gene-panel size	10	6.087	3.297	2.859	3.181	2.620	1.908	1.374	1.243	1.310	1.211
	50	1.995	1.561	1.318	1.747	1.233	0.969	0.890	0.819	0.920	0.794
	100	3.320	1.843	1.558	1.899	1.458	1.354	1.011	0.915	1.001	0.894

Beyond per-organ models, a single MIST trained jointly on all organs pooled (leave-slide-out cross-validation; no per-organ specialisation and no organ labels) predicts across the diverse organs at once, adapting to each slide through its shared local, global, and slide-level context rather than a dedicated per-tissue model. On the six-organ subset it attains the best mean PCC (0.599; Table 5), ahead of BLEEP (0.587) and Hist2ST (0.556), and reaches 0.760 on the full seven-organ HEST panel (BLEEP 0.758). Thus one shared model serves many organs at once, without per-organ retraining, provided each organ is represented during training.

Table 5: Cross-organ POOLED on the six-organ subset (kidney, breast, lung, pancreas, prostate, skin; COAD excluded). 5-fold pooled cross-validation (slide-level holdout, all organs seen); mean_{std} over three seeds; best in bold.

Method	PCC ↑
ST-Net He et al. (2020) (NBE’20)	0.351 _{0.02}
Hist2ST Zeng et al. (2022) (BIB’22)	0.556 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS’23)	0.587 _{0.00}
STEM Zhu et al. (2025) (ICLR’25)	0.273 _{0.00}
MIST	0.599_{0.01}

Table 2: Intra-cohort PCC across 50-gene selection strategies. Mean_{std} over three seeds.

Panel	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
HVG	IDC	0.551±0.01	0.305±0.11	0.519±0.00	0.434±0.00	0.581±0.01
	CCRCC	0.163±0.02	0.180±0.03	0.175±0.02	0.118±0.01	0.252±0.01
	LUNG	0.521±0.01	0.430±0.05	0.565±0.01	0.407±0.00	0.589±0.01
	PAAD	0.366±0.03	0.248±0.04	0.412±0.03	0.233±0.02	0.437±0.03
	PRAD	0.287±0.02	0.323±0.01	0.346±0.01	0.207±0.02	0.410±0.01
	SKCM	0.661±0.01	0.523±0.00	0.650±0.00	0.461±0.02	0.707±0.01
	Average	0.425	0.335	0.444	0.310	0.496
DEG	IDC	0.697±0.01	0.549±0.07	0.731±0.01	0.684±0.02	0.774±0.01
	CCRCC	0.312±0.01	0.335±0.01	0.304±0.00	0.253±0.01	0.402±0.01
	LUNG	0.566±0.01	0.518±0.01	0.621±0.01	0.440±0.01	0.662±0.00
	PAAD	0.425±0.03	0.352±0.01	0.487±0.03	0.306±0.04	0.518±0.03
	PRAD	0.413±0.00	0.437±0.01	0.466±0.00	0.355±0.03	0.539±0.01
	SKCM	0.752±0.01	0.656±0.00	0.746±0.03	0.617±0.02	0.794±0.00
	Average	0.528	0.475	0.559	0.443	0.615
HMHVG	IDC	0.724±0.01	0.633±0.08	0.739±0.02	0.687±0.01	0.774±0.01
	CCRCC	0.339±0.01	0.387±0.01	0.349±0.01	0.290±0.01	0.454±0.02
	LUNG	0.568±0.00	0.598±0.00	0.661±0.00	0.517±0.01	0.693±0.00
	PAAD	0.441±0.03	0.341±0.04	0.493±0.02	0.305±0.03	0.524±0.02
	PRAD	0.452±0.00	0.477±0.01	0.432±0.04	0.337±0.02	0.555±0.02
	SKCM	0.824±0.03	0.710±0.14	0.849±0.02	0.731±0.06	0.901±0.00
	Average	0.558	0.524	0.587	0.478	0.650

Table 3: Intra-cohort PCC across gene-panel sizes. Mean_{std} over three seeds; each panel reports its average.

Metric	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
PCC@10	IDC	0.666±0.01	0.636±0.01	0.635±0.02	0.591±0.02	0.708±0.00
	CCRCC	0.231±0.01	0.273±0.02	0.231±0.02	0.170±0.00	0.322±0.01
	LUNG	0.702±0.01	0.519±0.02	0.735±0.00	0.616±0.01	0.759±0.00
	PAAD	0.511±0.04	0.377±0.07	0.536±0.03	0.393±0.04	0.519±0.05
	PRAD	0.442±0.03	0.459±0.02	0.434±0.03	0.325±0.03	0.524±0.02
	SKCM	0.728±0.00	0.471±0.04	0.697±0.01	0.598±0.01	0.769±0.01
	Average	0.547	0.456	0.545	0.449	0.600
PCC@50	IDC	0.551±0.01	0.309±0.07	0.519±0.00	0.439±0.01	0.581±0.01
	CCRCC	0.163±0.02	0.184±0.02	0.175±0.02	0.123±0.00	0.252±0.01
	LUNG	0.518±0.01	0.470±0.01	0.562±0.01	0.423±0.03	0.589±0.01
	PAAD	0.366±0.03	0.248±0.04	0.412±0.03	0.233±0.02	0.437±0.03
	PRAD	0.294±0.02	0.333±0.02	0.348±0.01	0.207±0.02	0.410±0.01
	SKCM	0.665±0.01	0.522±0.03	0.645±0.01	0.491±0.04	0.707±0.01
	Average	0.425	0.335	0.444	0.310	0.496
PCC@100	IDC	0.611±0.01	0.473±0.07	0.632±0.01	0.565±0.01	0.682±0.01
	CCRCC	0.221±0.01	0.274±0.03	0.247±0.02	0.192±0.00 [†]	0.328±0.02
	LUNG	0.595±0.00	0.434±0.03	0.659±0.01	0.498±0.03	0.688±0.00
	PAAD	0.355±0.04	0.222±0.04	0.399±0.04	0.236±0.02	0.435±0.03
	PRAD	0.327±0.01	0.341±0.03	0.377±0.02	0.240±0.03	0.450±0.00
	SKCM	0.696±0.01	0.446±0.12	0.696±0.01	0.602±0.02	0.771±0.01
	Average	0.468	0.365	0.502	0.389	0.559

Key Finding (RQ1): MIST gives the best accuracy across diverse organs both per organ (intra-cohort PCC@50 0.496 vs. 0.444 for BLEEP, winning all six cohorts) and as a single pooled model (0.599 six-organ / 0.760 full HEST), so a separate model per tissue is unnecessary when the organ is represented in training.

5.2 RQ2: Cross-organ transfer to an unseen organ

Leave-one-organ-out (LOOO) is substantially harder than the pooled setting of RQ1 because the target anatomy is absent from training. On the six-organ subset MIST attains the best mean PCC (0.422; Table 11) and wins every held-out organ. On the full seven-organ HEST panel it reaches 0.490 versus 0.473 for BLEEP, the strongest baseline; BLEEP is stronger on IDC and SKCM there. Appendix H reports a complementary seven-organ mouse experiment.

Key Finding (RQ2): The pooled→LOOO drop—from 0.760 with the organ seen (RQ1) to 0.490 with the organ held out—isolates unseen-organ transfer as the dominant remaining gap; predicting seen organs well does not by itself transfer to a new organ.

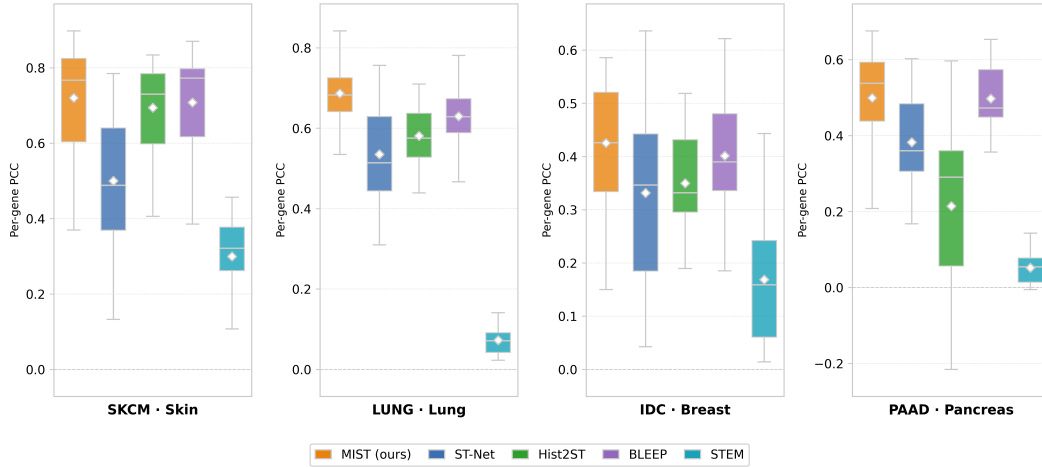


Figure 2: Per-cohort cross-organ LOOO accuracy (four of the six cohorts in Table 11; kidney and prostate omitted for readability, frozen UNI+CONCH). One panel per held-out cohort; box = per-gene PCC over three seeds, $\diamond$ = mean. MIST (orange) leads on all four.

5.3 RQ3: What drives accuracy and coordinate invariance?

The factorial study evaluates all eight combinations of local (L), global (G), and slide-level (S) context on SKCM and LUNG. Local attention provides the largest gain: enabling it alone raises mean PCC from 0.608 to 0.640. Global and slide context have little effect without the local pathway, but the complete model achieves the highest mean (0.643). This result supports distance-biased neighborhood aggregation as the primary contributor in the tested intra-cohort setting; claims about component importance under organ shift require a matched LOOO ablation not reported here.

Table 6 numerically verifies the analytic invariance result. Differences under translation and rotation remain below 6×10^{-5} and are attributable to floating-point evaluation; reflection is exact in this test.

Key Finding (RQ3): Local distance-biased attention is the primary driver of accuracy—enabling it alone lifts mean PCC from 0.608 to 0.640, while global and slide context add little without it—and the predictor is provably invariant to spot translation, rotation, and reflection.

Table 6: Numerical coordinate-invariance check. Maximum absolute prediction difference when only the spot coordinates are transformed (randomly initialized model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Intra-cohort spatial gene expression: Skin (SKCM), slide TENX117

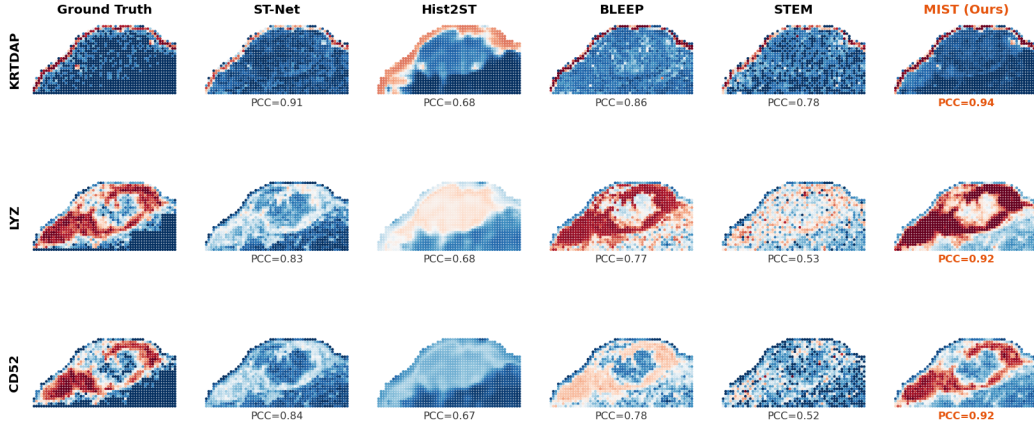


Figure 3: Qualitative spatial expression maps for a held-out skin (SKCM) slide (TENX117), three genes (rows). Each panel colors every spot by expression; MIST (right) most faithfully recovers the ground-truth spatial structure: the perivascular ring for LYZ/CD52 and the epidermal edge for KRTDAP. It attains the highest per-gene PCC (in orange), whereas STEM is noisy and Hist2ST over-smooths. Predictions use identical frozen UNI+CONCH features.

Table 7: Context-stream ablation (six-organ POOLED, no COAD). Each row toggles a subset of MIST’s local (L), global (G), and slide (S) context streams on the six-organ POOLED set (CCRCC/IDC/LUNG/PAAD/PRAD/SKCM; frozen UNI+CONCH, 14-gene, 5-fold). POOLED PCC $\uparrow$, MSE $\downarrow$, and MAE $\downarrow$ are mean $_{\text{std}}$ over three seeds; best per column in bold. The last row replaces spatial-kNN local attention with random neighbours. The complete model (L+G+S) is best on PCC and MSE; local attention is the dominant stream.

Configuration	Streams	PCC $\uparrow$	MSE $\downarrow$	MAE $\downarrow$
Global + Local	L + G	0.60 $\pm$ 0.01	0.46 $\pm$ 0.02	0.41 $\pm$ 0.01
Global + Slide	G + S	0.57 $\pm$ 0.04	0.52 $\pm$ 0.13	0.44 $\pm$ 0.05
Global only	G	0.59 $\pm$ 0.00	0.47 $\pm$ 0.02	0.42 $\pm$ 0.01
Local + Slide	L + S	0.60 $\pm$ 0.01	0.43 $\pm$ 0.03	0.40$\pm$0.02
Local only	L	0.60 $\pm$ 0.02	0.46 $\pm$ 0.05	0.41 $\pm$ 0.02
Slide only	S	0.58 $\pm$ 0.02	0.47 $\pm$ 0.03	0.42 $\pm$ 0.02
Global + Local (random nbr.)	L $_{\text{rand}}$ + G	0.58 $\pm$ 0.01	0.45 $\pm$ 0.01	0.40$\pm$0.00
Full, random nbr.	L $_{\text{rand}}$ + G + S	0.58 $\pm$ 0.01	0.47 $\pm$ 0.01	0.41 $\pm$ 0.00
MIST (ours)	L + G + S	0.61$\pm$0.01	0.42$\pm$0.01	0.40$\pm$0.01

5.4 RQ4: Spatial structure preservation

Gene-wise PCC does not test whether predictions preserve coherent tissue regions. We cluster spots (KMeans, $k=6$ domains) using either predicted or measured expression and compare the induced spatial partitions with adjusted Rand index (ARI) and normalized mutual information (NMI) on held-out slides of the six-organ subset (Table 8). MIST leads on both ARI (0.173) and NMI (0.242), ahead of the strongest baseline on each metric (BLEEP at 0.147 ARI, ST-Net at 0.201 NMI). The differences remain small and slide-level variation is large ($\text{std} \approx \text{mean}$), so we caution against over-interpreting them as evidence of uniformly improved domain recovery; the full seven-cohort version is in Appendix I.

Table 8: Spatial-domain recovery (six-organ subset). ARI/NMI between domains clustered (KMeans, $k=6$) from predicted vs. measured expression on held-out slides of CCRCC, IDC, LUNG, PAAD, PRAD, and SKCM (COAD excluded); mean_{std} over three seeds, higher is better, best per column in bold.

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.140 _{0.11}	0.201 _{0.12}
Hist2ST Zeng et al. (2022)	0.133 _{0.11}	0.180 _{0.11}
BLEEP Xie et al. (2023)	0.147 _{0.12}	0.197 _{0.13}
STEM Zhu et al. (2025)	0.101 _{0.12}	0.136 _{0.14}
MIST	0.173_{0.14}	0.242_{0.14}

5.5 RQ5: Computational efficiency

MIST contains 4.88M parameters and predicts a complete slide in one forward pass (Appendix Table 12). Its advantage over the baselines is not minimum parameter count: ST-Net and BLEEP are smaller, whereas MIST combines multi-scale spatial context with direct inference. Model-only latency grows from 3.5ms for 500 spots to 64.7ms for 10,000 spots on an A100, with 3.1 GiB peak allocated memory at the largest size (Appendix Table 13); feature extraction and disk I/O are excluded. STEM’s iterative diffusion inference is the only approach with substantially higher latency.

Three findings merit emphasis. First, a compact direct predictor is competitive when features, gene panels, splits, and model selection are controlled; the 10.3% relative gain over the strongest baseline rests on matched experimental conditions, not a larger model. Second, pooled multi-organ performance is not a substitute for unseen-organ evaluation: the drop from 0.760 (pooled) to 0.490 (LOOO) shows that anatomical shift remains a separate and harder problem. Third, local distance-biased attention is the dominant contributor in the factorial study; global and slide pathways provide moderate complementary gains, and the complete model (C111) achieves the best mean across both tested cohorts.

Direct and generative models serve different purposes. STEM [Zhu et al. \(2025\)](#) models a conditional expression distribution and can represent one-to-many uncertainty through iterative sampling. MIST produces a point estimate optimized for correlation and squared error. Its benefits, including deterministic, single-pass inference and explicit coordinate invariance, do not imply that morphology uniquely determines expression.

Limitations. The evaluation predicts 50 genes from frozen UNI+CONCH embeddings; results may not extend to transcriptome-scale output or end-to-end image encoding. LOOO uses a fixed benchmark panel and does not test platform or institution shift simultaneously. Coordinate invariance holds only for the spatial pathway with fixed image features; rotating the histology image can change its frozen embedding. Domain-recovery analysis covers only two baselines and has high slide-level variance. Pooled folds are sample-disjoint but cannot be verified as patient-disjoint where patient identifiers are unavailable.

6 Conclusion and Future Work

We developed MIST, a coordinate-invariant local-global spatial transformer for predicting gene expression from histology. Pairwise-distance encoding makes the spatial pathway invariant to rigid coordinate transformations by construction, while three parallel context signals: local k NN attention, global self-attention, and slide-level pooling, together produce complete-slide predictions in a single pass. These design choices directly address our design goals that multi-scale context is necessary and that architectural positional bias should not break under coordinate shift. Under matched HEST-1k evaluation, MIST improves mean intra-cohort correlation and remains competitive in LOOO transfer; the pooled-to-LOOO gap exposes the remaining difficulty of anatomical transfer as the central open challenge.

For future work, it would be valuable to explore how conditioning on slide-level metadata (organ type, sequencing platform) can narrow this anatomical gap, for instance, through FiLM or adaptive layer normalisation conditioners that are available at inference time without requiring expression labels. Scaling the gene panel beyond 50 genes, incorporating end-to-end image encoding, and combining MIST’s geometric structure with calibrated generative uncertainty modeling (as in STEM) are further natural directions.

AI use statement

Generative AI tools assisted with manuscript organization, language editing, ICLR formatting, and consistency checks between narrative claims and reported tables. They were not used to generate experimental data or execute model experiments. The authors reviewed all AI-assisted revisions and take responsibility for the final content of this work.

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Supplementary Material

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A Dataset

Table 9 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent cohorts differ substantially in size: IDC has $2.25\times$ more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has $2.2\times$ more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 3 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate. Our cross-organ experiments (leave-one-organ-out and pooled) use the six cohorts marked in Table 9: CCRCC (kidney, 24 samples / 74,220 spots), IDC (breast, 4 / 44,974), LUNG (lung, 2 / 7,505), PAAD (pancreas, 3 / 9,793), PRAD (prostate, 23 / 62,710), and SKCM (skin, 2 / 5,716); COAD, READ, LYMPH_IDC, and HCC are not used in the cross-organ setting.

Table 9: HEST-1k cohort statistics. Spots counted after barcode filtering. Folds = number of train/test splits in the per-cohort cross-validation. The six cohorts marked $\checkmark$ (CCRCC, IDC, LUNG, PAAD, PRAD, SKCM) form the no-COAD cross-organ evaluation set used in our leave-one-organ-out and pooled experiments; LYMPH_IDC, COAD, READ, and HCC (liver) are not used there (HCC has a near-zero signal floor).

Cohort	Organ	Samples	Spots	Folds	6-organ set
IDC	Breast	4	44,974	4	$\checkmark$
LYMPH_IDC	Breast	4	19,964	4	
COAD	Colorectal	4	18,523	2	
READ	Colorectal	4	8,407	2	
CCRCC	Kidney	24	74,220	6	$\checkmark$
HCC	Liver*	2	4,248	2	
LUNG	Lung	2	7,505	2	$\checkmark$
PAAD	Pancreas	3	9,793	3	$\checkmark$
PRAD	Prostate	23	62,710	2	$\checkmark$
SKCM	Skin	2	5,716	2	$\checkmark$

*HCC (liver) is included here for completeness but excluded from Table 3; all baselines score near zero on this cohort due to low spatial gene-expression variance.

B Extended Related Work

Machine-learning prediction of gene expression from histology. ST-Net He et al. (2020) established patch-level regression. Hist2ST Zeng et al. (2022) adds spatial context through a graph over neighboring spots, while BLEEP Xie et al. (2023) aligns image and expression embeddings for cross-modal retrieval. STEM Zhu et al. (2025) instead treats expression prediction as conditional generation. These methods use different features, gene panels, splits, and selection procedures, so their published scores do not isolate architectural gains; our comparisons therefore use a matched protocol. MIST is a deterministic point predictor and does not model the full conditional expression distribution.

Geometric structure and coordinate invariance. Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this by constructing interactions from relative quantities Satorras et al. (2021). MIST uses pairwise Euclidean distances, which are preserved by translations, rotations, and reflections; the guarantee applies to the spatial pathway for fixed histology features, not to rotations of the image-encoder input.

Generalization across samples and organs. HEST-1k [Jaume et al. \(2024\)](#) spans many cohorts and organs. Within-cohort evaluation can exploit organ-specific morphology or acquisition effects. We therefore distinguish pooled cross-validation, which holds out samples while retaining their organ types in training, from LOOO, which withholds an entire organ and provides a stricter test of anatomical transfer.

Positioning of MIST. MIST connects these three directions: an inductive-bias-explicit spatial transformer for direct regression, evaluated under a matched cross-cohort protocol. Unlike absolute position encodings or frame-averaging strategies, its coordinate pathway is invariant by construction. Unlike generative models, it does not represent expression uncertainty; its advantage is deterministic, single-pass inference with explicit geometric structure. Unlike prior evaluations that report within-cohort accuracy only, our controlled comparison separates interpolation from anatomical transfer, making the difficulty of organ shift directly measurable.

C Extended Experimental Setup

Dataset and preprocessing. We evaluate on HEST-1k [Jaume et al. \(2024\)](#) using the cohorts of Table 9; HCC, LYMPH_IDC, and READ are excluded under the documented quality and split criteria, and the six cohorts marked in that table (CCRCC, IDC, LUNG, PAAD, PRAD, SKCM) form the cross-organ set. We use the provided patient-stratified folds, $\log(1+x)$ normalization, and frozen UNI+CONCH features (1536-d). Unless stated otherwise, targets are the 50-gene HEST HVG panel. The panel study additionally uses matched 50-gene DEG (Wilcoxon selection across Leiden clusters) and HMMHVG (mean-and-variability selection after variance filtering) panels (Table 2).

Evaluation protocol. Intra-cohort training fits a model within each cohort. Pooled five-fold validation trains across organs while holding out slide-level samples; organ identity defines the protocol but is not a model input. LOOO trains on all-but-one organ and tests the held-out organ. We report PCC, SCC, and MSE averaged over outer folds and three seeds. Checkpoints are selected on inner validation and evaluated once on the outer test fold; single-slide folds use disjoint spots for inner validation. Patient-level disjointness is not guaranteed where identifiers are unavailable.

Baselines and implementation. We compare with ST-Net [He et al. \(2020\)](#), Hist2ST [Zeng et al. \(2022\)](#), BLEEP [Xie et al. \(2023\)](#), and STEM [Zhu et al. \(2025\)](#) under matched features, panels, splits, budgets, and checkpoint selection. MCToGene uses a different ten-cohort protocol and appears in Appendix K. MIST uses four layers, width 256, four heads, eight neighbors, 16 radial bases, and 4.88M parameters. AdamW training uses learning rate 10^{-3} , weight decay 10^{-2} , cosine decay, gradient clipping, 100 epochs maximum, patience 20, and at most 3,000 spots per step.

D Computation Resources and Inference Tools

All experiments ran on a shared GPU cluster. Feature extraction and training used NVIDIA GPUs with compute capability $\geq \text{sm_80}$ (A100 80GB, A100-PCIE, L40s, and H200); Volta (V100) GPUs are incompatible with our CUDA-13 / PyTorch build and were excluded.

- Encoders: frozen UNI (ViT-L/16, 1024-d) and CONCH (512-d) patch features, concatenated to 1536-d; no encoder fine-tuning.
- Model: MIST has 4.88M trainable parameters (Table 12) and predicts a whole slide in a single forward pass—no iterative sampling.
- Training: AdamW with a cosine schedule, 100 epochs with early stopping (patience 20), nested validation, and a correlation-aware objective; three seeds per configuration.
- Inference: single-pass; median model-only latency and peak memory versus slide size are reported in Table 13, with feature extraction and disk I/O excluded.

E Encoder Ablation

We repeat the intra-cohort PCC@50 comparison (the 50-gene block of Table 3, which uses the concatenated UNI + CONCH features) with each foundation encoder in isolation: UNI only (ViT-L/16, 1024-d) and CONCH only (512-d), holding models, cohorts, gene panels, splits, and training budget fixed (Table 10). — entries are still training.

Table 10: Encoder ablation: intra-cohort PCC@50 with each foundation encoder in isolation. UNI only (1024-d) vs. CONCH only (512-d), per cohort; mean_{std} over three seeds; best per row in bold. Compare with the UNI + CONCH (1536-d) results in Table 3.

Encoder	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
UNI (1024-d)	IDC Breast	0.538±0.008	0.250±0.067	0.508±0.005	—	0.569±0.009
	CCRCC Kidney	0.172±0.010	0.214±0.015	0.184±0.007	—	0.264±0.010
	LUNG Lung	0.482±0.012	0.378±0.097	0.559±0.007	0.393±0.009	0.590±0.005
	PAAD Pancreas	0.353±0.022	0.243±0.061	0.393±0.035	—	0.392±0.040
	PRAD Prostate	0.251±0.008	0.322±0.020	0.343±0.009	—	0.394±0.003
	SKCM Skin	0.632±0.010	0.510±0.015	0.639±0.015	0.471±0.031	0.694±0.001
	Average	0.405	0.319	0.438	—	0.484
CONCH (512-d)	IDC Breast	0.515±0.007	0.264±0.018	0.496±0.011	0.396±0.011	0.551±0.004
	CCRCC Kidney	0.160±0.010	0.126±0.026	0.141±0.012	—	0.222±0.010
	LUNG Lung	0.517±0.001	0.435±0.006	0.576±0.003	0.386±0.030	0.554±0.005
	PAAD Pancreas	0.372±0.035	0.186±0.035	0.384±0.033	0.251±0.028	0.387±0.045
	PRAD Prostate	0.327±0.003	0.322±0.009	0.364±0.010	0.210±0.013	0.406±0.006
	SKCM Skin	0.633±0.002	0.388±0.022	0.633±0.005	0.453±0.014	0.658±0.007
	Average	0.421	0.287	0.432	—	0.463

F Complete LOOO (Leave-One-Organ-Out) Results

Table 11 reports the complete per-cohort leave-one-organ-out transfer results on the six-organ subset summarised in the main text (Section 5.2, Figure 2). Each fold holds out one entire organ and trains on the other five, using leakage-safe per-fold gene panels and identical frozen UNI + CONCH features. MIST wins every held-out cohort and the average (0.422).

Table 11: Cross-organ LOOO transfer on a six-organ subset (kidney, breast, lung, pancreas, prostate, skin; COAD excluded), per-cohort Pearson $\uparrow$. Self-contained leave-one-organ-out: each fold holds out one organ and trains on the other five, with leakage-safe per-fold gene panels. Identical frozen UNI + CONCH features (1536-dim); mean_{std} over three seeds; best per row in bold. MIST wins every cohort and the average.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25 [†]	MIST (Ours)
CCRCC Kidney	0.074±0.00	0.064±0.01	0.058±0.00	0.024±0.01	0.098±0.01
IDC Breast	0.332±0.02	0.350±0.00	0.401±0.04	0.131±0.08	0.425±0.03
LUNG Lung	0.535±0.07	0.581±0.00	0.630±0.00	0.067±0.00	0.687±0.02
PAAD Pancreas	0.382±0.00	0.214±0.17	0.497±0.03	0.052±0.00	0.499±0.03
PRAD Prostate	0.078±0.01	0.062±0.01	0.091±0.00	0.016±0.00	0.105±0.01
SKCM Skin	0.500±0.09	0.694±0.02	0.708±0.01	0.299±0.00	0.720±0.03
Average	0.317	0.328	0.398	0.098	0.422

[†]STEM incomplete (1–2 of 3 seeds); values provisional.

G Model complexity and inference scaling

Table 12: Model complexity. Parameter counts and inference mechanisms.

Model	#Params (M)	Inference
ST-Net He et al. (2020) (NBE’20)	0.47	single-pass
Hist2ST Zeng et al. (2022) (BIB’22)	2.18	single-pass
BLEEP Xie et al. (2023) (NeurIPS’23)	0.54	single-pass
STEM Zhu et al. (2025) (ICLR’25)	34.16	iterative
MIST (V3)	4.88	single-pass

Table 13: MIST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

H Mouse cross-organ diagnostic

We evaluate 59 mouse Visium slides spanning seven organs with mouse-specific 50-gene panels. Pooled performance reaches 0.887, whereas every method remains near the floor under LOOO; we therefore do not interpret the latter as evidence of successful cross-species organ transfer.

Table 14: Cross-organ generalization on mouse HEST (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: five-fold pooled cross-validation. PCC $\uparrow$ on 50 mouse HVGs.

Method	looo PCC †	pooled PCC
ST-Net He et al. (2020)	0.013 _{0.011}	0.357 _{0.036}
Hist2ST Zeng et al. (2022)	0.018 _{0.007}	0.829 _{0.028}
BLEEP Xie et al. (2023)	-0.006 _{0.003}	0.848 _{0.012}
STEM Zhu et al. (2025)	-	0.279 _{0.009}
MIST	0.018_{0.006}	0.887_{0.005}

† All looo values are near the floor because the organ-specific gene panels do not transfer reliably.

I Spatial-domain recovery

Table 15: Spatial-domain recovery (all seven cohorts). ARI/NMI between domains clustered (KMeans, $k=6$) from predicted vs. measured expression on held-out slides of all seven retained HEST-1k cohorts; mean_{std} over three seeds, higher is better, best per column in bold. Companion to the five-organ Table 8.

Method	ARI $\uparrow$	NMI $\uparrow$
ST-Net He et al. (2020)	0.149 _{0.11}	0.213 _{0.13}
Hist2ST Zeng et al. (2022)	0.149 _{0.12}	0.197 _{0.12}
BLEEP Xie et al. (2023)	0.157 _{0.12}	0.209 _{0.14}
STEM Zhu et al. (2025)	0.113 _{0.13}	0.152 _{0.15}
MIST	0.192_{0.15}	0.259_{0.16}

J Cross-organ LOOO transfer with UNI features

Table 16 reports a complementary LOOO comparison using UNI features. EGN has the highest mean PCC (0.493), while MIST reaches 0.477. M2OST narrowly leads EGN on LUNG (0.667 versus 0.666). STEM uses the corrected DiT configuration with $T = 1000$; its SKCM entry contains one completed seed and is marked accordingly.

Table 16: Cross-organ LOOO transfer with UNI features (Pearson mean_{std} over three seeds; nested-validation; HCC, LYMPH_IDC, and READ excluded). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	MIST (V3)
IDC Breast	0.217 $\pm$ 0.06	0.459 $\pm$ 0.02	0.509 $\pm$ 0.03	0.452 $\pm$ 0.01	0.531 $\pm$ 0.03	0.406 $\pm$ 0.01	0.534$\pm$0.02
COAD Colorectal	0.430 $\pm$ 0.03	0.688 $\pm$ 0.04	0.667 $\pm$ 0.02	0.672 $\pm$ 0.03	0.749$\pm$0.01	0.504 $\pm$ 0.03	0.703 $\pm$ 0.01
CCRCC Kidney	0.084 $\pm$ 0.03	0.084 $\pm$ 0.01	0.100 $\pm$ 0.02	0.122 $\pm$ 0.01	0.106 $\pm$ 0.01	0.090 $\pm$ 0.01	0.124$\pm$0.00
LUNG Lung	0.504 $\pm$ 0.04	0.499 $\pm$ 0.06	0.621 $\pm$ 0.02	0.667$\pm$0.01	0.666 $\pm$ 0.01	0.561 $\pm$ 0.01	0.660 $\pm$ 0.01
PAAD Pancreas	0.404 $\pm$ 0.05	0.360 $\pm$ 0.07	0.521 $\pm$ 0.02	0.522 $\pm$ 0.01	0.551$\pm$0.02	0.460 $\pm$ 0.02	0.507 $\pm$ 0.04
PRAD Prostate	0.071 $\pm$ 0.01	0.104 $\pm$ 0.02	0.111 $\pm$ 0.00	0.107 $\pm$ 0.02	0.098 $\pm$ 0.02	0.068 $\pm$ 0.01	0.119$\pm$0.02
SKCM Skin	0.402 $\pm$ 0.08	0.652 $\pm$ 0.08	0.745 $\pm$ 0.01	0.724 $\pm$ 0.02	0.749$\pm$0.01	0.686 [†]	0.692 $\pm$ 0.06
Average	0.302	0.407	0.468	0.466	0.493	0.396 [†]	0.477

[†]STEM DiT (corrected, T=1000, 100 epochs) LOOO: SKCM (skin) has only 1 seed complete; all other organs use 3 seeds. Average marked [†] accordingly.

K Published Per-Cohort Comparison

Table 17: Published per-cohort accuracy on the standard HEST-1k splits, mean PCC $\uparrow$ as reported in respective papers (ten cohorts, no organ grouping). Shown for context only (not comparable to Table 3), which uses organ grouping and nested-validation. [†]MIST uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net He et al. (2020) (NBE'20)	0.286
Hist2ST Zeng et al. (2022) (BIB'22)	–
MCToGene Sun et al. (2026) (CVPR'26)	0.435
MIST [†] (ours)	0.440

L Gene Panel Comparison

Table 18 reports intra-cohort mean PCC averaged over seven cohorts and three seeds for three gene-panel strategies: highly variable genes (HVG, the HEST-1k default), unsupervised differentially expressed genes (DEG; Leiden clustering + Wilcoxon), and high-mean highly variable genes (HMHVG; selected using expression mean and variability after variance filtering). The supplied panel-generation script constructs DEG and HMHVG panels once per cohort using all available slides, rather than selecting genes separately within each training fold. These results therefore describe an exploratory comparison with fixed cohort-level panels; held-out expression can influence target selection, and nested check-point validation does not remove this dependence. A strictly inductive comparison requires training-fold-only panel selection and retraining. All methods use identical UNI+CONCH features, nested-validation, and the same training protocol. DEG and HMHVG panels consistently outperform HVG across all methods, with HMHVG yielding the highest mean for every method tested. MIST leads on all three panels, with its advantage most pronounced on HMHVG (0.657 vs. 0.595 for BLEEP) and consistent across DEG (0.624 vs. 0.571) and HVG (0.471 vs. 0.427).

Table 18: Gene panel comparison: intra-cohort mean PCC $\uparrow$ (HEST-1k). Mean_{std} over three seeds; per-seed value is the average over seven cohorts. All methods use identical frozen UNI + CONCH features (1536-dim) and nested-validation. Best per column in bold.

Method	HVG	DEG	HMHVG
ST-Net He et al. (2020)	0.398 _{0.004}	0.523 _{0.004}	0.550 _{0.008}
Hist2ST Zeng et al. (2022)	0.330 _{0.032}	0.494 _{0.014}	0.544 _{0.038}
BLEEP Xie et al. (2023)	0.427 _{0.004}	0.571 _{0.005}	0.595 _{0.014}
STEM Zhu et al. (2025)	0.291 _{0.005}	0.456 _{0.002}	0.485 _{0.010}
MIST (ours)	0.471_{0.005}	0.624_{0.002}	0.657_{0.006}

M Proofs of the formal guarantees

M.1 Proof of Proposition 1

For every pair of spots i, j , orthogonality of R gives

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|R(\mathbf{p}_i - \mathbf{p}_j)\|_2 \quad (9)$$

$$= \|\mathbf{p}_i - \mathbf{p}_j\|_2. \quad (10)$$

Hence, transforming the coordinate frame leaves the complete distance matrix unchanged. It also leaves the k NN graph, radial basis features, and distance-dependent attention biases unchanged. The input projection is unchanged because $\mathbf{F}$ is fixed. At every layer, the local pathway therefore receives the same features, neighbors, and distance biases, while the global and slide pathways do not use coordinates. Their outputs, the residual fusion, and the feed-forward update are identical. Induction over the L layers gives identical final representations and predictions, proving Proposition 1.

M.2 Proof of Proposition 2

Let $\mathbf{F}' = Q\mathbf{F}$ and $\mathbf{P}' = Q\mathbf{P}$. The pairwise distance matrix is relabelled as $\mathbf{D}' = Q\mathbf{D}Q^\top$. Under equivariant tie handling, the k NN graph and its radial basis edge features are relabelled by the same permutation. The input projection and all pointwise linear maps, layer normalizations, activations, and feed-forward networks commute with Q .

For local attention, permuting the spots only relabels each neighborhood and its attention sum, so the local output is multiplied by Q . Global self-attention is likewise permutation equivariant because its query-key score matrix is transformed by Q on both axes and its value matrix by Q . The slide attention scores are permuted, but their weighted sum is unchanged; broadcasting that same summary to all spots is equivariant. Consequently, residual fusion preserves equivariance at every layer. Induction over the layers and application of the pointwise prediction head yield $f_\theta(Q\mathbf{F}, Q\mathbf{P}) = Qf_\theta(\mathbf{F}, \mathbf{P})$, proving Proposition 2.

N Three-Metric Intra-Cohort Summary

Table 19 extends the intra-cohort comparison with Spearman correlation (SCC) and mean squared error (MSE) alongside PCC, averaged over seven organ groups.

Table 19: Intra-cohort three-metric summary (HEST-1k, averaged over seven organ groups; three seeds; UNI + CONCH features). PCC $\uparrow$, SCC $\uparrow$, MSE $\downarrow$. Best per column in bold.

Method	PCC $\uparrow$	SCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE'20)	0.399	0.356	2.043
Hist2ST Zeng et al. (2022) (BIB'22)	0.337	0.321	1.616
BLEEP Xie et al. (2023) (NeurIPS'23)	0.426	0.388	1.464
STEM Zhu et al. (2025) (ICLR'25)	0.299	0.274	1.855
MIST (ours)	0.470	0.430	1.389

O Per-Cohort MSE and MAE (Intra-Cohort)

Tables 20 and 21 report the per-cohort mean squared error (MSE) and mean absolute error (MAE) for the same models, cohorts, and protocol as the intra-cohort Pearson comparison in Table 3. MIST attains the best average on both error metrics (MSE 1.389, MAE 0.829), consistent with its Pearson result; per-cohort leadership is less uniform than on Pearson, as the training objective couples MSE with a correlation term.

Table 20: Intra-cohort MSE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 3.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST (Ours)
IDC Breast	2.966 $\pm$ 0.311	2.740 $\pm$ 0.194	2.503 $\pm$ 0.061	2.811 $\pm$ 0.056	2.242$\pm$0.031
COAD Colorectal	2.335 $\pm$ 0.063	2.195$\pm$0.058	2.201 $\pm$ 0.062	2.632 $\pm$ 0.021	2.322 $\pm$ 0.031
CCRCC Kidney	0.587 $\pm$ 0.007	0.578 $\pm$ 0.032	0.604 $\pm$ 0.011	0.763 $\pm$ 0.016	0.556$\pm$0.019
LUNG Lung	3.388 $\pm$ 0.062	1.732 $\pm$ 0.003	1.650$\pm$0.024	2.036 $\pm$ 0.052	1.665 $\pm$ 0.043
PAAD Pancreas	2.051 $\pm$ 0.235	1.331 $\pm$ 0.193	1.151$\pm$0.258	1.711 $\pm$ 0.206	1.173 $\pm$ 0.166
PRAD Prostate	0.586 $\pm$ 0.011	0.560 $\pm$ 0.020	0.570 $\pm$ 0.019	0.822 $\pm$ 0.014	0.537$\pm$0.034
SKCM Skin	2.387 $\pm$ 0.085	2.176 $\pm$ 0.016	1.570 $\pm$ 0.036	2.207 $\pm$ 0.260	1.227$\pm$0.030
Average	2.043	1.616	1.464	1.855	1.389

Table 21: Intra-cohort MAE $\downarrow$ on HEST-1k, per cohort. Mean_{std} over three seeds with nested-validation; best (lowest) per row in bold. Same models, cohorts, features (frozen UNI + CONCH, 1536-dim), 50-gene panels, and training budget as Table 3.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	BLEEP NeurIPS'23	STEM ICLR'25	MIST (Ours)
IDC Breast	1.273 $\pm$ 0.066	1.246 $\pm$ 0.032	1.141 $\pm$ 0.022	1.210 $\pm$ 0.015	1.096$\pm$0.010
COAD Colorectal	0.996 $\pm$ 0.014	1.008 $\pm$ 0.017	0.986$\pm$0.020	1.088 $\pm$ 0.008	1.040 $\pm$ 0.012
CCRCC Kidney	0.575 $\pm$ 0.003	0.574 $\pm$ 0.018	0.583 $\pm$ 0.003	0.612 $\pm$ 0.011	0.569$\pm$0.008
LUNG Lung	1.373 $\pm$ 0.014	0.985 $\pm$ 0.002	0.960 $\pm$ 0.008	1.050 $\pm$ 0.003	0.957$\pm$0.014
PAAD Pancreas	1.013 $\pm$ 0.051	0.872 $\pm$ 0.055	0.796$\pm$0.088	0.936 $\pm$ 0.054	0.798 $\pm$ 0.046
PRAD Prostate	0.564 $\pm$ 0.004	0.546$\pm$0.003	0.547 $\pm$ 0.007	0.620 $\pm$ 0.014	0.550 $\pm$ 0.017
SKCM Skin	1.015 $\pm$ 0.023	1.056 $\pm$ 0.015	0.926 $\pm$ 0.011	1.049 $\pm$ 0.070	0.791$\pm$0.005
Average	0.973	0.898	0.848	0.938	0.829

P Per-Cohort Breakdown of the Error Metrics (Table 4)

Table 4 reports MSE and MAE averaged over the six cohorts for each gene-selection strategy and panel size. Table 22 (MSE across selection strategies) and Table 23 (MAE across panel sizes) give the per-cohort values behind those averages: the mean of each column equals the corresponding entry of Table 4. Same six cohorts, five models, frozen UNI + CONCH features, splits, and training budget as the Pearson Tables 2 and 3; mean_{std} over three seeds; best (lowest) per row in bold.

Table 22: Intra-cohort MSE $\downarrow$ across 50-gene selection strategies, per cohort.

Panel	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
HVG	IDC Breast	2.98 $\pm$ 0.28	2.86 $\pm$ 0.39	2.43 $\pm$ 0.04	2.80 $\pm$ 0.06	2.24$\pm$0.03
	CCRCC Kidney	0.59 $\pm$ 0.01	0.57 $\pm$ 0.01	0.60 $\pm$ 0.01	0.80 $\pm$ 0.04	0.55$\pm$0.02
	LUNG Lung	3.32 $\pm$ 0.03	1.80 $\pm$ 0.06	1.62$\pm$0.04	2.10 $\pm$ 0.09	1.67 $\pm$ 0.04
	PAAD Pancreas	2.04 $\pm$ 0.27	1.34 $\pm$ 0.21	1.13$\pm$0.26	1.71 $\pm$ 0.21	1.17 $\pm$ 0.17
	PRAD Prostate	0.63 $\pm$ 0.03	0.56 $\pm$ 0.00	0.57 $\pm$ 0.02	0.82 $\pm$ 0.01	0.54$\pm$0.03
	SKCM Skin	2.41 $\pm$ 0.08	2.24 $\pm$ 0.08	1.55 $\pm$ 0.05	2.25 $\pm$ 0.17	1.23$\pm$0.03
	Average	0.425	0.335	0.444	0.310	0.496
DEG	IDC Breast	5.42 $\pm$ 0.32	2.55 $\pm$ 0.83	1.93 $\pm$ 0.24	2.04 $\pm$ 0.28	1.85$\pm$0.17
	CCRCC Kidney	2.10 $\pm$ 0.16	1.58 $\pm$ 0.01	1.67 $\pm$ 0.01	1.90 $\pm$ 0.06	1.58$\pm$0.10
	LUNG Lung	3.87 $\pm$ 0.04	1.95 $\pm$ 0.01	1.87$\pm$0.01	2.45 $\pm$ 0.05	1.92 $\pm$ 0.02
	PAAD Pancreas	3.35 $\pm$ 0.38	1.46 $\pm$ 0.24	1.32$\pm$0.29	1.97 $\pm$ 0.33	1.34 $\pm$ 0.24
	PRAD Prostate	2.18 $\pm$ 0.31	1.55 $\pm$ 0.03	1.48 $\pm$ 0.03	2.01 $\pm$ 0.06	1.34$\pm$0.03
	SKCM Skin	3.09 $\pm$ 0.09	1.45 $\pm$ 0.03	1.08 $\pm$ 0.10	1.37 $\pm$ 0.13	0.83$\pm$0.01
	Average	0.528	0.475	0.559	0.443	0.615
HMHVG	IDC Breast	4.72 $\pm$ 0.60	2.24 $\pm$ 0.46	2.09 $\pm$ 0.28	2.28 $\pm$ 0.13	1.80$\pm$0.15
	CCRCC Kidney	1.46 $\pm$ 0.04	1.24 $\pm$ 0.08	1.23 $\pm$ 0.02	1.41 $\pm$ 0.02	1.12$\pm$0.09
	LUNG Lung	4.55 $\pm$ 0.02	1.89 $\pm$ 0.02	1.68 $\pm$ 0.04	2.05 $\pm$ 0.00	1.60$\pm$0.01
	PAAD Pancreas	2.58 $\pm$ 0.26	1.37 $\pm$ 0.28	1.21$\pm$0.23	1.76 $\pm$ 0.22	1.24 $\pm$ 0.19
	PRAD Prostate	1.52 $\pm$ 0.15	1.35$\pm$0.02	1.40 $\pm$ 0.07	1.89 $\pm$ 0.15	1.52 $\pm$ 0.35
	SKCM Skin	6.71 $\pm$ 0.41	2.90 $\pm$ 0.91	1.80 $\pm$ 0.26	2.47 $\pm$ 0.64	1.10$\pm$0.03
	Average	0.558	0.524	0.587	0.478	0.650

Table 23: Intra-cohort MAE $\downarrow$ across gene-panel sizes, per cohort.

Genes	Cohort	ST-Net	Hist2ST	BLEEP	STEM	MIST (Ours)
10	IDC Breast	2.87 $\pm$ 0.21	1.79 $\pm$ 0.12	1.75$\pm$0.02	1.83 $\pm$ 0.04	1.77 $\pm$ 0.11
	CCRCC Kidney	0.78 $\pm$ 0.01	0.74 $\pm$ 0.01	0.74 $\pm$ 0.01	0.79 $\pm$ 0.01	0.74$\pm$0.02
	LUNG Lung	2.55 $\pm$ 0.05	1.62 $\pm$ 0.02	1.34 $\pm$ 0.02	1.41 $\pm$ 0.06	1.26$\pm$0.01
	PAAD Pancreas	2.22 $\pm$ 0.16	1.41 $\pm$ 0.21	1.25$\pm$0.21	1.36 $\pm$ 0.20	1.33 $\pm$ 0.28
	PRAD Prostate	0.94 $\pm$ 0.05	0.83 $\pm$ 0.01	0.82$\pm$0.01	0.86 $\pm$ 0.01	0.83 $\pm$ 0.02
	SKCM Skin	2.09 $\pm$ 0.02	1.85 $\pm$ 0.10	1.56 $\pm$ 0.00	1.61 $\pm$ 0.06	1.34$\pm$0.03
	Average	1.28 $\pm$ 0.06	1.27 $\pm$ 0.09	1.12 $\pm$ 0.01	1.21 $\pm$ 0.02	1.10$\pm$0.01
50	IDC Breast	0.58 $\pm$ 0.00	0.57 $\pm$ 0.00	0.58 $\pm$ 0.00	0.63 $\pm$ 0.02	0.57$\pm$0.01
	CCRCC Kidney	1.36 $\pm$ 0.01	1.01 $\pm$ 0.02	0.95$\pm$0.01	1.06 $\pm$ 0.03	0.96 $\pm$ 0.01
	LUNG Lung	1.01 $\pm$ 0.06	0.87 $\pm$ 0.07	0.79$\pm$0.09	0.94 $\pm$ 0.05	0.80 $\pm$ 0.05
	PAAD Pancreas	0.57 $\pm$ 0.02	0.56 $\pm$ 0.01	0.55$\pm$0.01	0.62 $\pm$ 0.01	0.55 $\pm$ 0.02
	PRAD Prostate	1.02 $\pm$ 0.02	1.06 $\pm$ 0.02	0.92 $\pm$ 0.02	1.06 $\pm$ 0.04	0.79$\pm$0.00
	SKCM Skin	1.66 $\pm$ 0.05	1.15 $\pm$ 0.05	1.05$\pm$0.06	1.16 $\pm$ 0.08	1.06 $\pm$ 0.02
	Average	0.91 $\pm$ 0.02	0.80 $\pm$ 0.02	0.82 $\pm$ 0.00	0.88 $\pm$ 0.01	0.80$\pm$0.00
100	IDC Breast	1.71 $\pm$ 0.01	1.05 $\pm$ 0.01	0.89$\pm$0.00	1.01 $\pm$ 0.01	0.91 $\pm$ 0.01
	CCRCC Kidney	0.97 $\pm$ 0.06	0.80 $\pm$ 0.07	0.74$\pm$0.06	0.87 $\pm$ 0.05	0.75 $\pm$ 0.04
	LUNG Lung	0.94 $\pm$ 0.05	0.91 $\pm$ 0.02	0.87$\pm$0.01	0.94 $\pm$ 0.02	0.92 $\pm$ 0.09
	PAAD Pancreas	1.93 $\pm$ 0.02	1.36 $\pm$ 0.06	1.13 $\pm$ 0.01	1.15 $\pm$ 0.01	0.92$\pm$0.01
	PRAD Prostate					
	SKCM Skin					
	Average					

Q Cross-Organ Generalization: PCC and MSE

Table 24 reports both PCC and MSE for the LOOO and POOLED cross-organ regimes, complementing the PCC-only summaries in the main paper.

Table 24: Cross-organ generalization: PCC and MSE (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: 5-fold pooled cross-validation. PCC $\uparrow$ / MSE $\downarrow$ on HEST-1k.

Method	looo		pooled	
	PCC $\uparrow$	MSE $\downarrow$	PCC $\uparrow$	MSE $\downarrow$
ST-Net He et al. (2020) (NBE’20)	0.267 _{0.02}	1.846 _{0.10}	0.370 _{0.01}	0.907 _{0.05}
Hist2ST Zeng et al. (2022) (BIB’22)	0.360 _{0.02}	1.182 _{0.07}	0.737 _{0.01}	0.488 _{0.01}
BLEEP Xie et al. (2023) (NeurIPS’23)	0.416 _{0.01}	1.010_{0.02}	0.758 _{0.01}	0.427_{0.02}
STEM Zhu et al. (2025) (ICLR’25)	0.322 _{0.01}	1.274 _{0.00}	0.643 _{0.01}	0.594 _{0.01}
MIST (ours)	0.421_{0.01}	1.233 _{0.11}	0.760_{0.01}	0.440 _{0.02}

R Intra-Cohort Results with UNI + CONCH Features, Extended Baselines

Table 25 extends the intra-cohort comparison to include M2OST and EGN under the same UNI + CONCH feature protocol.

Table 25: Intra-cohort results using UNI + CONCH features, extended baselines (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix A. Best per row in bold.

Cohort	ST-Net NBE’20	Hist2ST BIB’22	BLEEP NeurIPS’23	M2OST AAAI’25	EGN WACV’23	STEM ICLR’25	MIST (ours)
IDC Breast	0.551 $\pm$ 0.01	0.309 $\pm$ 0.06	0.519 $\pm$ 0.00	0.571 $\pm$ 0.01	0.551 $\pm$ 0.01	0.439 $\pm$ 0.01	0.581$\pm$0.01
COAD Colorectal	0.242 $\pm$ 0.01	0.295 $\pm$ 0.01	0.314 $\pm$ 0.01	0.328$\pm$0.01	0.293 $\pm$ 0.01	0.177 $\pm$ 0.01	0.318 $\pm$ 0.01
CCRCC Kidney	0.163 $\pm$ 0.02	0.184 $\pm$ 0.02	0.175 $\pm$ 0.02	0.210 $\pm$ 0.02	0.223 $\pm$ 0.02	0.123 $\pm$ 0.00	0.252$\pm$0.01
LUNG Lung	0.518 $\pm$ 0.01	0.470 $\pm$ 0.01	0.562 $\pm$ 0.01	0.598$\pm$0.00	0.584 $\pm$ 0.01	0.423 $\pm$ 0.03	0.589 $\pm$ 0.01
PAAD Pancreas	0.359 $\pm$ 0.03	0.247 $\pm$ 0.02	0.419 $\pm$ 0.03	0.413 $\pm$ 0.04	0.415 $\pm$ 0.03	0.233 $\pm$ 0.02	0.437$\pm$0.03
PRAD Prostate	0.294 $\pm$ 0.02	0.333 $\pm$ 0.02	0.348 $\pm$ 0.01	0.413$\pm$0.01	0.377 $\pm$ 0.03	0.207 $\pm$ 0.02	0.410 $\pm$ 0.01
SKCM Skin	0.665 $\pm$ 0.01	0.522 $\pm$ 0.03	0.645 $\pm$ 0.01	0.654 $\pm$ 0.01	0.631 $\pm$ 0.01	0.491 $\pm$ 0.04	0.707$\pm$0.01
Average	0.399	0.337	0.426	0.455	0.439	0.299	0.470

All entries use three completed seeds. STEM uses the corrected DiT configuration ($T = 1000$, 100 epochs).

Table 26: Intra-cohort PCC@50 with UNI-only features (1024-d). Same six cohorts, models, and protocol as Table 3; $\text{mean} \pm \text{std}$ over three seeds; best per row in bold. Only completed runs are shown; “_” denotes a result not available in the workspace.

Cohort	ST-Net NBE’20	Hist2ST BIB’22	HyperST CVPR’26	BLEEP NeurIPS’23	STEM ICLR’25	PatchMorphoST (ours)
IDC Breast	—	—	—	—	—	0.569 $\pm$ 0.011
COAD Colorectal	—	—	—	—	—	—
CCRCC Kidney	—	—	—	—	—	0.264 $\pm$ 0.012
LUNG Lung	—	—	—	—	—	0.590 $\pm$ 0.007
PAAD Pancreas	—	—	—	—	—	0.392 $\pm$ 0.049
PRAD Prostate	—	—	—	—	—	0.394 $\pm$ 0.003
SKCM Skin	—	—	—	—	—	0.694 $\pm$ 0.001
Average of available rows	—	—	—	—	—	0.484

UNI-only PatchMorphoST results are from results_pcc50_uni_morphost; COAD is absent from the completed UNI run. No baseline UNI-only results were found.

S Training Objective Ablation

Table 27 isolates the contribution of the Pearson correlation term in the training objective from the architectural design, confirming that the accuracy gap over baselines is not attributable to the loss function alone.

Table 27: Objective ablation (HEST-1k). Effect of adding the Pearson correlation term to the MSE objective; averaged over seven organ groups, three seeds.

Model	Objective	PCC $\uparrow$
MIST	MSE only	0.382
MIST	MSE + PCC	0.389